



EPFL AUTOMATIC LABORATORY
DDMAC

PROFESSOR : ALIREZA KARIMI // PHD ASSISTANT : VAIBHAV GUPTA

Semester project report

MICROGRIDS PRIMARY - SECONDARY DISTRIBUTED CONTROL WITH DELAY
UNCERTAINTY

1	Introduction	1
2	Grid model	2
2.1	Component models	2
2.1.1	Loads	2
2.1.2	Photovoltaic panels	3
2.1.3	Batteries	3
2.1.4	Synchronous generator	4
2.2	Line Power Flows model	4
2.3	Complete microgrid model	5
2.4	Synchronous frequency control	6
3	System Identification	7
3.1	MIMO Identification	7
3.2	Results	7
4	Control design	9
4.1	Control specifications	9
4.2	Linear Fractional representations	9
4.2.1	Frequency performance	9
4.2.2	Power sharing	10
4.2.3	Mixed sensitivity	12
4.3	Controller design by convex optimization	12
4.4	Communication delays	13
4.4.1	Fixed terms in the controller	13
4.4.2	Delay uncertainty	14
4.5	Results	14
4.5.1	Distributed Data-Driven controller	14
4.5.2	Distributed Data-Driven controller with delay multimodel uncertainty	15
5	Conclusion	18

List of Figures

1	Microgrid schema	2
2	3 nodes grid schema	5
3	synchronous frequency feedback	6
4	Schema of the close loop identification	7
5	Plant frequency response from frequency input ω_5 to active power P_5	8
6	Plant frequency response from frequency input ω_8 to active power P_8	8
7	Linear Fractional Representation Schema	9
8	Block diagram for sensitivity inputs performances	10
9	Weighting filter W_3^{-1} Bode Diagram	10
10	Block diagram for sensitivity loop-shaping performances	11
11	Weighting filter α^{-1} Bode Diagram	12
12	Block diagram for mixed sensitivity performances	12
13	Comparison of frequency performances between the droop controller 'blue' and Data-Driven distributed controller 'red' after impulse 'left' and step perturbation 'right' . .	14
14	Comparison of power sharing performances between the droop controller 'blue' and Data-Driven distributed controller 'red'	15
15	Comparison of frequency performances between the droop controller 'blue', incorrect delay Data-Driven controller 'red' and the multimodel delay data-driven controller . .	16
16	Comparison of power sharing performances between the droop controller 'blue', incorrect delay Data-Driven controller 'red' and the multimodel delay data-driven controller . .	17

List of Tables

1	Loads nominal power consumption	3
2	Photovoltaic panels nominal power generation	3
3	Batteries nominal power generation	3
4	Synchronous generator nominal power generation	4
5	Distributed controller frequency performances	15
6	Distributed controller power sharing performances	15
7	Multimodel controller frequency performances	16
8	Multimodel controller power sharing performances	17

1 Introduction

In the rapidly evolving landscape of energy systems, microgrids have emerged as a pivotal component, offering decentralized and resilient solutions for power generation and distribution. The effectiveness of a microgrid hinges on the precision and adaptability of its control system, which plays a crucial role in ensuring stability, frequency regulation, and optimal power sharing among diverse energy sources.

This report delves into a comprehensive exploration of microgrid control, leveraging a data-driven approach to design an optimized controller. With an emphasis on frequency stability and power-sharing performance, our study investigates the intricacies of controller design, incorporating elements such as communication delays and uncertainties. This work aims to explore deeper the one establish in [1]. Through simulation experiments and comparative analyses, we aim to unveil the nuanced dynamics of microgrid control systems and assess the efficacy of the proposed data-driven controller.

The report unfolds with an overview of microgrid components and control objectives, setting the stage for a detailed examination of the linear fractional transformation (LFT) framework. We explore how this framework accommodates communication delays and uncertainties, shedding light on the integration of these factors into the optimization problem.

Subsequently, the report presents a meticulous analysis of the performance metrics, emphasizing the controller's ability to maintain grid frequency stability and optimize power sharing among distributed energy resources. Real-world simulation results are juxtaposed with those from an idealized scenario, providing valuable insights into the controller's robustness and adaptability.

2 Grid model

In this section, our initial focus is on introducing microgrids and their components. Subsequently, we aim to construct a linear model to gain a comprehensive understanding of the control objectives. A microgrid functioning in islanded mode comprises four types of nodes, as detailed in this section. These include the load, responsible for energy consumption, and distributed generators (such as solar panels) that generate energy but are susceptible to external perturbations (such as weather conditions). In instances where solar panels do not produce the required power, our control nodes, namely batteries and synchronous generators, come into play to meet the power demand. The electrical connections linking these nodes consist of both resistive and inductive elements.

Here is the schematic representation of an example microgrid operating in islanded mode (Figure 1). It comprises four loads (nodes 3, 4, 6, and 9), three photovoltaic systems (nodes 1, 2, and 7), one synchronous generator (node 5), and two batteries (nodes 8 and 10). The nodes under control are highlighted in red, and we presume communication links between them, as depicted by the red arrows. This microgrid is the one studied in this overall experiment.

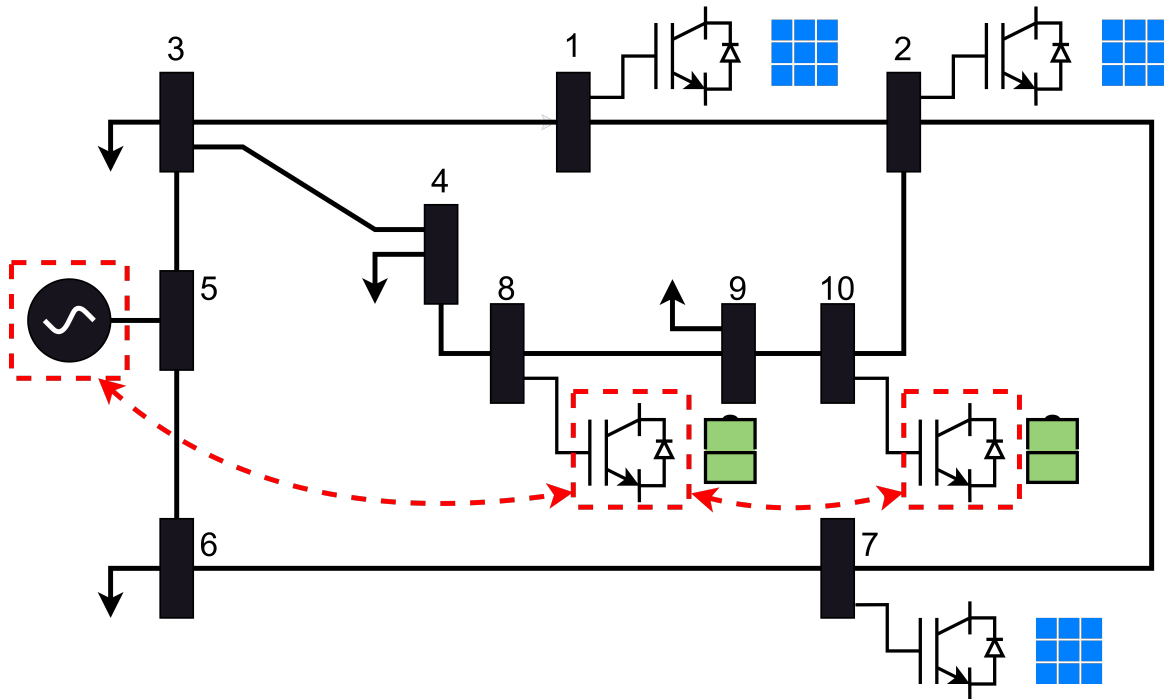


Figure 1: Microgrid schema

2.1 Component models

We now aim to understand the control inputs and outputs of our entire system. Initially, the subsystem models will be outlined, providing a detailed description of each. Subsequently, we will present the integrated model of the overall system.

2.1.1 Loads

The loads represent the most straightforward components of the grid. We make the assumption that their power consumption remains quasi-constant (active and reactive). Any alteration in load power

consumption is regarded as a perturbation within the grid.

In the studied grid, the nominal power consumption of the loads are presented in the following table :

Bus	Active Power [kW]	Reactive Power [kW]
3	30	0
4	20	0
6	25	0
9	45	0

Table 1: Loads nominal power consumption

2.1.2 Photovoltaic panels

Photovoltaic panels generally produce direct current (DC), while the microgrid functions on alternating current (AC). In order to align the DC electricity generated by photovoltaic panels with the grid's voltage and frequency, inverters are employed for the conversion process. A current controller is established to guarantee that the power injected into the grid falls within acceptable limits.

In the context of short-term experiments, it is reasonable to assume constant weather conditions, resulting in consistent power generation summarized in the following table :

Bus	Active Power [kW]	Reactive Power [kW]
1	10	0
2	10	0
7	10	0

Table 2: Photovoltaic panels nominal power generation

2.1.3 Batteries

Our first control subsystems involve the batteries. Similar to photovoltaic panels, batteries belong to the category of distributed generators, generating continuous voltage (DC). To convert this continuous voltage into alternating signal (AC), an inverter is employed. The system's inputs encompass the desired frequency and voltage, while the outputs comprise the actual frequency and voltages generated by the inverter.

Voltage Source Inverters (VSIs) can be represented by a first-order dynamic equation with respective time constants τ_ω and τ_U for frequency and voltage. The corresponding transfer function is then expressed as follows:

$$\begin{bmatrix} \omega_I \\ U_I \end{bmatrix} = G_I \cdot \begin{bmatrix} \bar{\omega}_I \\ \bar{U}_I \end{bmatrix}, \quad \text{with } G_I = \begin{bmatrix} \frac{1}{\tau_\omega s + 1} & 0 \\ 0 & \frac{1}{\tau_U s + 1} \end{bmatrix} \quad (1)$$

The nominal power generation of each of the batteries is summarized in the table below :

Bus	Active Power [kW]	Reactive Power [kW]
8	35	0
10	20	0

Table 3: Batteries nominal power generation

2.1.4 Synchronous generator

A synchronous generator transforms mechanical energy into electricity at a constant speed synchronized with the microgrid's frequency. The generator frequency is directly correlated with the normalized difference between the mechanical and electrical powers, expressed as:

$$2H \frac{\dot{\omega}_S}{\omega_S^n} = \frac{P_{S,m} - P_S}{P_S^n} \quad (2)$$

The control of the synchronous generator involves manipulating the mechanical power. The transfer function between the desired mechanical power and the actual power is assumed to exhibit first-order dynamics with a time constant τ_m . The linearized model of the synchronous generator from power to frequency can thus be expressed as follows:

$$\omega_S = [G_{S,m} \quad G_{S,e}] \cdot \begin{bmatrix} \bar{P}_{S,m} \\ -P_S \end{bmatrix}, \quad \text{with } G_{S,m} = \frac{1}{\tau_m s + 1} \cdot \frac{\omega_S^n}{2HP_S^n s} \text{ and } G_{S,e} = \frac{\omega_S^n}{2HP_S^n s} \quad (3)$$

The voltage control can be expressed in a simplified manner. We make the assumption that the transfer function from the desired voltage to the actual voltage follows first-order dynamics with a time constant τ_U .

$$U_S = G_{S,U} \bar{U}_S \quad \text{with } G_{S,U} = \frac{1}{\tau_U s + 1} \quad (4)$$

The nominal power generation of each of the batteries is summarized in the table below :

Bus	Active Power [kW]	Reactive Power [kW]
5	35	0

Table 4: Synchronous generator nominal power generation

2.2 Line Power Flows model

To construct the model for line power flows, a dynamic phasor model is employed, as comprehensively detailed in the reference [2]. This modeling technique facilitates the expression of power at one node based on the frequency and voltage parameters of another interconnected node. This dynamic phasor model is particularly useful in understanding the complex interactions and power exchanges between different nodes in the microgrid. The linearized model of the line power flows is the following :

$$G_{i,j}^{\omega \rightarrow P}(s) = 3 \frac{\omega^n L_{i,j}}{(L_{i,j}s + R_{i,j})^2 + (\omega^n L_{i,j})^2} U^{n^2} \frac{1}{s} \quad (5)$$

$$G_{i,j}^{U \rightarrow P}(s) = 3 \frac{L_{i,j}s + R_{i,j}}{(L_{i,j}s + R_{i,j})^2 + (\omega^n L_{i,j})^2} U^n \quad (6)$$

$$G_{i,j}^{\omega \rightarrow Q}(s) = 3 \frac{L_{i,j}s + R_{i,j}}{(L_{i,j}s + R_{i,j})^2 + (\omega^n L_{i,j})^2} U^{n^2} \frac{1}{s} \quad (7)$$

$$G_{i,j}^{U \rightarrow Q}(s) = 3 \frac{\omega^n L_{i,j}}{(L_{i,j}s + R_{i,j})^2 + (\omega^n L_{i,j})^2} U^n \quad (8)$$

The comprehensive line transfer function of the grid can be computed as:

$$\begin{bmatrix} P_G \\ Q_G \\ P_L \\ Q_L \end{bmatrix} = \underbrace{\begin{bmatrix} G_1 & G_2 \\ G_3 & G_4 \end{bmatrix}}_{G_{line}} \cdot \begin{bmatrix} \omega_G \\ U_G \\ \omega_L \\ U_L \end{bmatrix} \quad (9)$$

Where $P_G//Q_G$ are the power generated from generator nodes and $P_L//Q_L$ are the power consumed by the nodes.

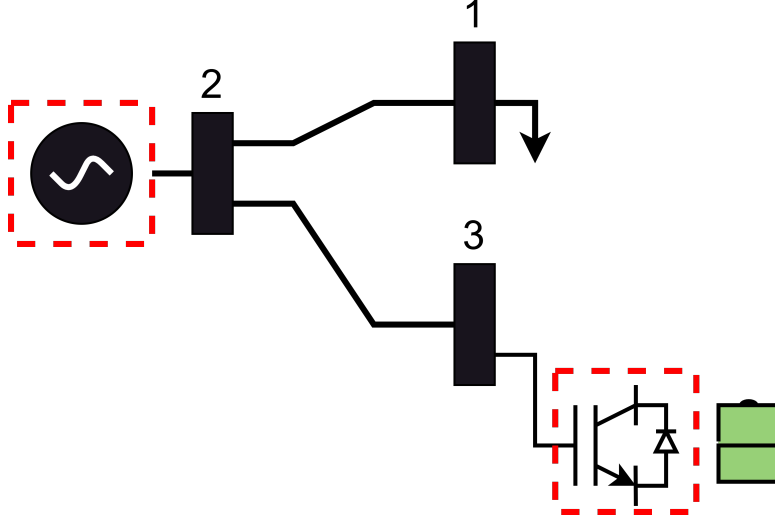


Figure 2: 3 nodes grid schema

In the simplified schema shown in the figure above (Figure 2) we have two generator nodes and one load. hence we can define G_1 as :

$$G_1 = \begin{bmatrix} G_{21}^{\omega \rightarrow P} + G_{23}^{\omega \rightarrow P} & G_{21}^{U \rightarrow P} + G_{23}^{U \rightarrow P} \\ G_{32}^{\omega \rightarrow P} & G_{32}^{U \rightarrow P} \\ G_{21}^{\omega \rightarrow Q} + G_{23}^{\omega \rightarrow Q} & G_{21}^{U \rightarrow Q} + G_{23}^{U \rightarrow Q} \\ G_{32}^{\omega \rightarrow Q} & G_{32}^{U \rightarrow Q} \end{bmatrix} \quad (10)$$

As previously mentioned, changes in load power consumption are considered perturbations in the system, and they are beyond our control. Therefore, the model needs to distinguish between controllable powers and perturbation powers. To achieve this, the matrix G_4 is inverted, and given the fully connected nature of the grid, it is always invertible. The resulting grid model is then as follows:

$$\begin{bmatrix} P_C \\ Q_C \end{bmatrix} = G_{grid} \cdot \begin{bmatrix} \omega_C \\ U_C \end{bmatrix} + G_d \cdot \begin{bmatrix} P_L \\ Q_L \end{bmatrix}, \quad \text{with } \begin{cases} G_{grid} = G_1 - G_2 G_4^{-1} G_3 \\ G_d = G_2 G_4^{-1} \end{cases}. \quad (11)$$

2.3 Complete microgrid model

Now, we need to integrate the grid model discussed in the previous section with the subsystem models presented in section 2.1. Given the established models, we can denote G_{comp} as the transfer function from desired values to controllable ones.

$$\begin{bmatrix} P_I \\ Q_I \\ P_S \\ Q_S \\ \omega_S \end{bmatrix} = G_{comp} \begin{bmatrix} \bar{u} \\ \bar{P}_{S,m} \end{bmatrix}, \quad \text{with } \bar{u} = \begin{bmatrix} \bar{\omega}_I \\ \bar{U}_I \\ \bar{U}_S \end{bmatrix} \quad (12)$$

The computation of G_{comp} is directly inspired by the methodology outlined in [2], and the details are elucidated in the following equations.

$$G_{comp} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \\ G_{31} & G_{32} \\ G_{41} & G_{42} \\ G_{51} & G_{52} \end{bmatrix} \cdot \begin{bmatrix} G_I & 0 & 0 \\ 0 & G_{S,U} & 0 \\ 0 & 0 & I \end{bmatrix}, \quad \text{with} \quad \begin{cases} G_{11} = G_{grid}^{u \rightarrow P_I} - G_{grid}^{\omega_S \rightarrow P_I} G_{S,e} G_{31} \\ G_{12} = G_{grid}^{\omega_S \rightarrow P_I} (G_{S,m} - G_{S,e} G_{32}) \\ G_{21} = G_{grid}^{u \rightarrow Q_I} - G_{grid}^{\omega_S \rightarrow Q_I} G_{S,e} G_{31} \\ G_{22} = G_{grid}^{\omega_S \rightarrow Q_I} (G_{S,m} - G_{S,e} G_{32}) \\ G_{31} = (I + G_{grid}^{\omega_S \rightarrow P_S} G_{S,e})^{-1} G_{grid}^{u \rightarrow P_S} \\ G_{32} = (I + G_{grid}^{\omega_S \rightarrow P_S} G_{S,e})^{-1} G_{grid}^{\omega_S \rightarrow P_S} G_{S,m} \\ G_{41} = G_{grid}^{u \rightarrow Q_S} - G_{grid}^{\omega_S \rightarrow Q_S} G_{S,e} G_{31} \\ G_{42} = G_{grid}^{\omega_S \rightarrow Q_S} (G_{S,m} - G_{S,e} G_{32}) \\ G_{51} = G_{S,e} G_{31} \\ G_{52} = G_{S,m} - G_{S,e} G_{32} \end{cases} \quad (13)$$

2.4 Synchronous frequency control

As observed, up to this point, the synchronous generator frequency is controllable but is not considered one of the system inputs. To address this, a closed-loop control is applied to this signal, aiming to derive a comprehensive model of the grid that encompasses the relationships from frequency and voltage to active and reactive powers.

A schematic representation of the final linear model is depicted below:

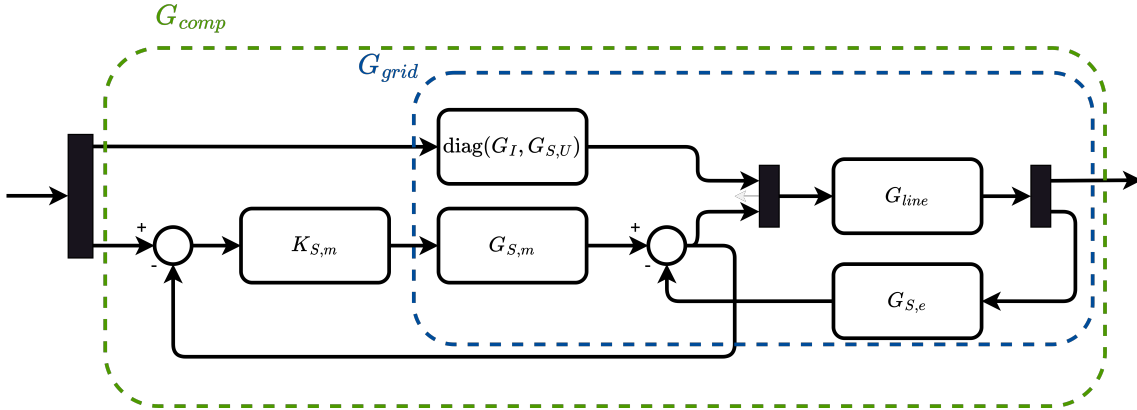


Figure 3: synchronous frequency feedback

3 System Identification

In practical systems, numerous parameters are subject to uncertainty, and a linear model may be too simplistic to accurately represent the grid dynamics. An alternative approach involves explicitly identifying the grid frequency response around a nominal point. The objective of this section is to outline a method for identifying the grid frequency response and subsequently compare it with the previously designed linear model.

3.1 MIMO Identification

In summary, the goal is to identify a Multi-Input Multi-Output (MIMO) system around a nominal point, which corresponds to the nominal power of all the controlled subsystems. The process begins with the implementation of a droop stabilizing controller to stabilize the system around this nominal point. Subsequently, a closed-loop MIMO identification is conducted.

To simplify the design and avoid some numerical issues, the inputs and outputs of the plant G are normalized to $U_{\text{base}} = 230 \text{ V}$, $\omega_{\text{base}} = 2\pi \cdot 50 \text{ rad/s}$, and $S_{\text{base}} = 100 \text{ kVA}$. A schema of the close loop applied to the normalized system is shown below :

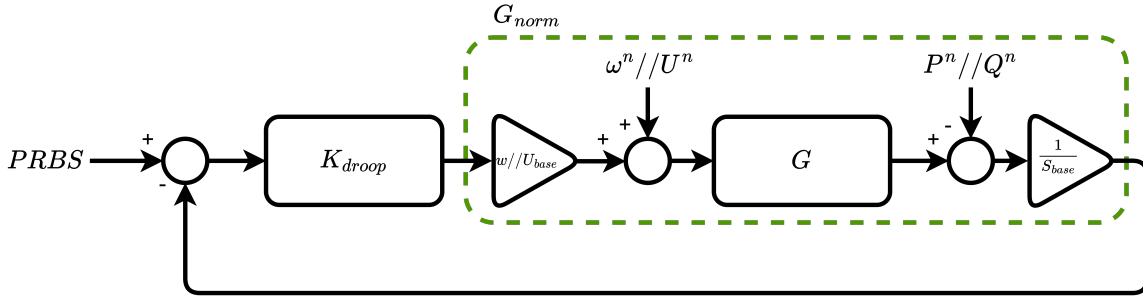


Figure 4: Schema of the close loop identification

A Pseudo-Random Binary Sequence (PRBS) reference signal $r(k)$ is applied to only one entry (indexed as "i") of the system. The resulting temporal input and output responses of the plant are captured in matrices $u_i(k)$ and $y_i(k)$, respectively. The frequency response of the inputs $U_i(e^{j\omega_k})$ and outputs $T_i(e^{j\omega_k})$ with respect to the PRBS reference can then be computed at specific frequency points.

By repeating this process for each entry, we can construct the frequency response matrices for both the input and output signals:

$$\begin{aligned} U(e^{j\omega_k}) &= [U_1(e^{j\omega_k}), \dots, U_n(e^{j\omega_k})] \\ T(e^{j\omega_k}) &= [T_1(e^{j\omega_k}), \dots, T_n(e^{j\omega_k})] \end{aligned} \quad (14)$$

Finally, the frequency response of the normalized plant can be computed as:

$$G(e^{j\omega_k}) = T(e^{j\omega_k})/U(e^{j\omega_k}). \quad (15)$$

3.2 Results

Here, various subparts of the identified model G_{norm} are presented using different methods. All simulations were conducted, with the blue line representing ideal inverters and the red line representing

switching inverters. These are then compared with the linear model developed in the first section, represented by the yellow dashed line.

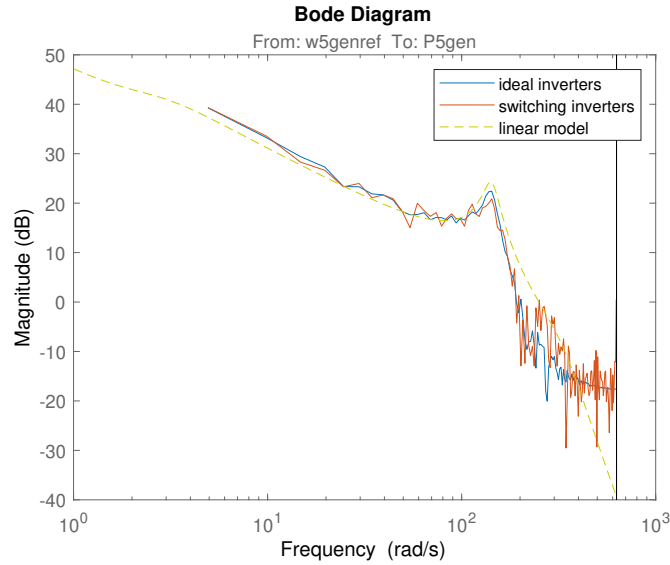


Figure 5: Plant frequency response from frequency input ω_5 to active power P_5

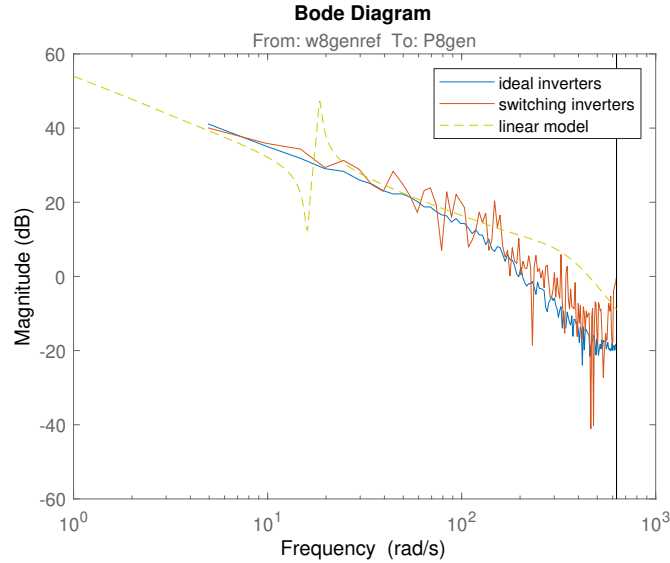


Figure 6: Plant frequency response from frequency input ω_8 to active power P_8

From these two figures, we observe remarkably similar results among the three curves. Although the linear model exhibits a potential issue around 20 rad/s (likely stemming from zero/pole cancellation in a matrix inversion due to numerical considerations), the overall agreement is notable. Therefore, for the remainder of the experiment, ideal inverters in simulation will be utilized to obtain an accurate model while significantly reducing simulation time compared to using switching inverters.

4 Control design

4.1 Control specifications

At this stage, we have successfully identified the frequency response of the plant, with the control inputs being the frequency and voltage, and the outputs being the active and reactive power for the controlled nodes only. We now want to understand what are the control specifications for a microgrid.

In the realm of microgrid control, the primary goal revolves around ensuring the stability and precise regulation of the grid frequency, typically set around 50Hz. Achieving stability is essential for consistent power supply, with control strategies in place to prevent oscillations and maintain system equilibrium under various operational conditions. Simultaneously, microgrid controllers aim to efficiently distribute generated power among diverse energy resources, adhering to their nominal capacities. This strategic power allocation not only optimizes the utilization of distributed generators but also prevents overloading or underutilization.

4.2 Linear Fractional representations

To derive the optimal controller, we employ an optimization approach based on data-driven optimization theory, as outlined in [3]. This involves the description of Linear Fractional Transformation (LFT) components associated with the desired specifications. The LFT framework provides a structured representation of the system that facilitates the incorporation of specific performance and stability requirements into the controller design process. By defining these LFT-related specifications, we can formulate the optimization problem, guiding the synthesis of a controller that meets the specified criteria and enhances the overall performance of the microgrid control system.

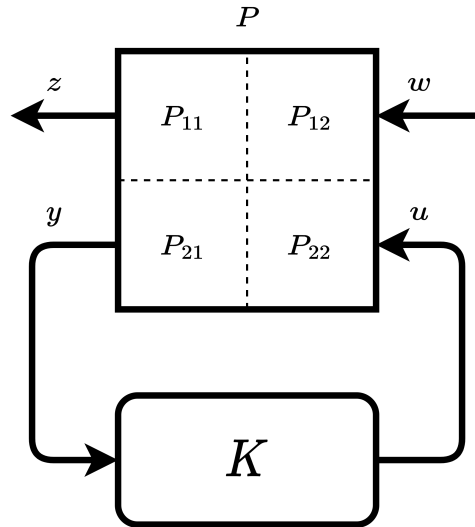


Figure 7: Linear Fractional Representation Schema

4.2.1 Frequency performance

To address frequency performance considerations in the optimization process, we begin by normalizing the plant around a frequency of 50Hz and a voltage of 230V. The objective is to minimize the control

inputs, ensuring that the system remains closely aligned with these normalized values. A block diagram is utilized to illustrate the interplay between the disturbance w_3 and the performance channel z_3 , providing a visual representation of the factors influencing frequency-related performance in the microgrid control system.

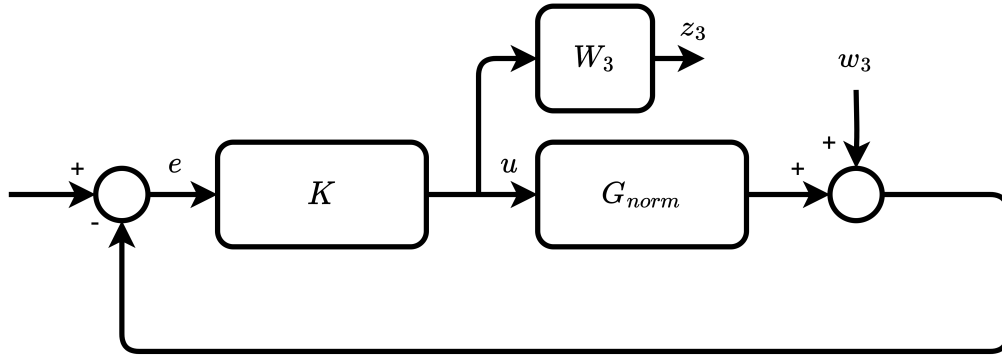


Figure 8: Block diagram for sensitivity inputs performances

From this diagram, the construction of the P matrix for the Linear Fractional Transformation (LFT) associated with this specific performance is straightforward. The weighting filter W_3 plays a key role in shaping the performance specifications and is defined as follows:

$$P_3 = \begin{bmatrix} 0 & W_3 \\ -I & -G \end{bmatrix} \quad (16)$$

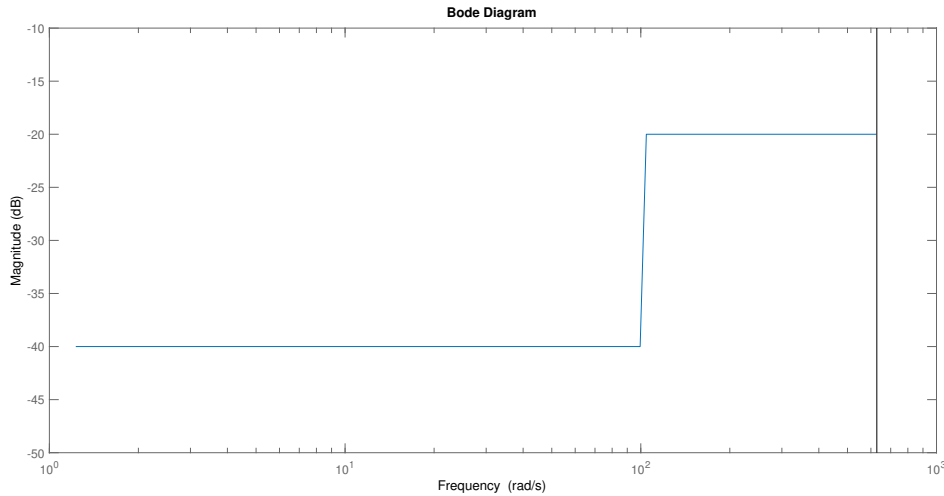


Figure 9: Weighting filter W_3^{-1} Bode Diagram

4.2.2 Power sharing

To tackle the power sharing challenge, we acknowledge the inherent difficulty of rejecting perturbations, as the total power consumption must be met. However, one objective is to effectively distribute the

power production among the various controlled generators. A loop-shaping design is employed for the sensitivity performance function, as illustrated in the following diagram (Figure 10).

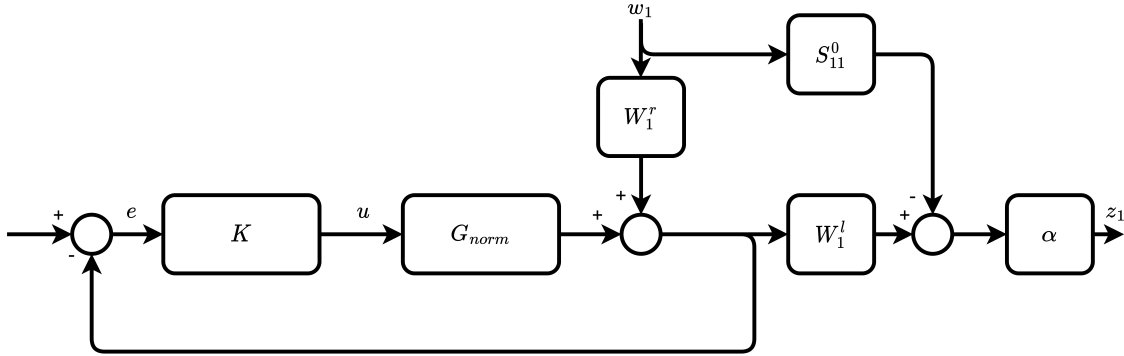


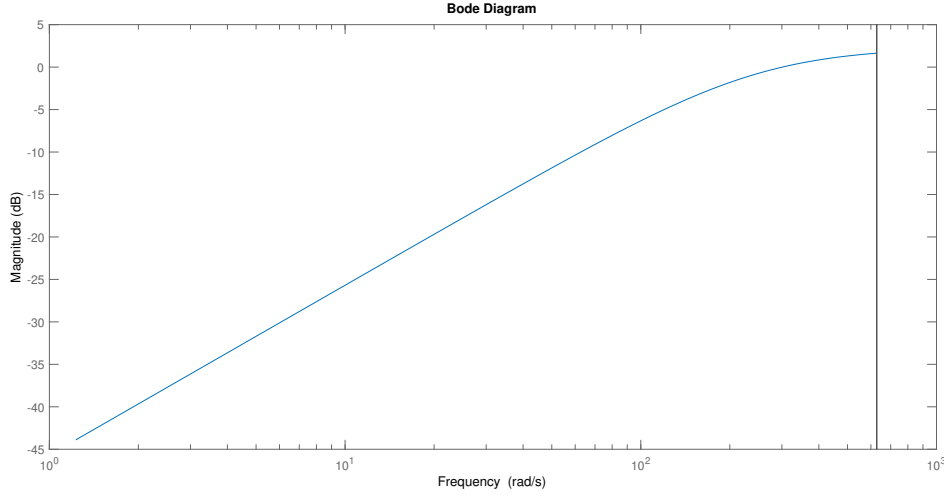
Figure 10: Block diagram for sensitivity loop-shaping performances

Based on the depicted diagram, the P matrix in the linear fractional transformation can be expressed as follows:

$$P_1 = \begin{bmatrix} \alpha(W_1^l W_1^r - [S_0^{11} \ 0]) & \alpha W_1^l G \\ -W_1^r & -G \end{bmatrix} \quad (17)$$

The weighting filters W_1^l and W_1^r serve the purpose of isolating the active power components of the sensitivity function. To ensure invertibility, small additional terms $\epsilon = 1e-2$ are incorporated into W_1^r . Lastly, the weighting filter α is tailored to assign greater significance to low frequencies. Its inverse, α^{-1} , is typically structured as a high-pass filter, as illustrated in the figure below.

$$\alpha = 100 \cdot \frac{50 + s}{1.3s}, \quad W_1^l = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad W_1^r = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \epsilon & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \epsilon & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \epsilon \end{bmatrix} \quad (18)$$

Figure 11: Weighting filter α^{-1} Bode Diagram

4.2.3 Mixed sensitivity

An alternative approach to consolidate both objectives into a single linear fractional transformation is to introduce both perturbations w_1 and w_2 into the same block diagram. This integrated representation allows for a unified treatment of the performance and power-sharing objectives within a single framework, providing a holistic view of the system dynamics and control interactions.

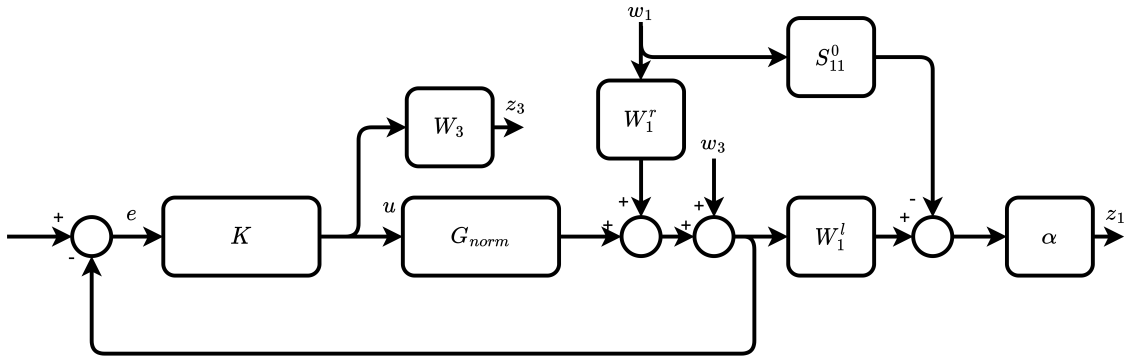


Figure 12: Block diagram for mixed sensitivity performances

The matrix P is then computed as follow :

$$P = \begin{bmatrix} \alpha(W_1^l W_1^r - [S_{11}^{11} \ 0]) & \alpha W_1^l G \\ 0 & W_3 \\ \begin{pmatrix} -W_1^r & 0 \\ 0 & -I \end{pmatrix} & -G \end{bmatrix} \quad (19)$$

4.3 Controller design by convex optimization

Now that we have computed the linear fractional transformations of the objective performances, we can proceed to formulate a convex optimization problem. This optimization problem will allow us to

systematically determine the controller parameters that optimize the overall system performance, balancing the specified frequency response and power-sharing objectives. The use of convex optimization techniques ensures a tractable and efficient solution approach.

To establish the transfer function between disturbances w and performance channels z , we begin by employing a right factorization for the controller, denoted as $K = XY^{-1}$. Within the equations, variables of the optimization process are highlighted in red.

$$\begin{aligned}
T_{z \leftarrow w} &= P_{11} + P_{12} \mathbf{K} (I - P_{22} \mathbf{K})^{-1} P_{21}, & (\mathbf{K} = \mathbf{X} \mathbf{Y}^{-1}) \\
&= P_{11} + P_{12} \mathbf{X} (\mathbf{Y} - P_{22} \mathbf{X})^{-1} P_{21}, & (\mathbf{\Phi} = P_{21}^R (\mathbf{Y} - P_{22} \mathbf{X})) \\
&= P_{11} + P_{12} \mathbf{X} \mathbf{\Phi}^L, & (\mathbf{\Psi} = I - \mathbf{\Phi} \mathbf{\Phi}^L = I - P_{21}^R P_{21}) \\
&= P_{11} (\underbrace{\mathbf{\Phi} \mathbf{\Phi}^L + \mathbf{\Psi}}_I) + P_{12} \mathbf{X} \mathbf{\Phi}^L \\
&= (P_{11} \mathbf{\Phi} + P_{12} \mathbf{X}) \mathbf{\Phi}^L + P_{11} \mathbf{\Psi}
\end{aligned} \tag{20}$$

Minimize the 2-norm of $T_{z \leftarrow w}$ is synonym of minimize trace $\int T_{z \leftarrow w} T_{z \leftarrow w}^* d\omega$. Which is decomposed as follow :

$$\begin{aligned}
\min \quad & \text{trace} \sum_{\omega \in \Omega} \Gamma(\omega) \cdot \Delta\omega \\
\text{s.t.} \quad & T_{z \leftarrow w} T_{z \leftarrow w}^* \preceq \Gamma(\omega), \quad (\forall \omega \in \Omega)
\end{aligned} \tag{21}$$

We want to write the constraints as LMIs in order to define a semi-definite program.

$$\begin{aligned}
& T_{z \leftarrow w} T_{z \leftarrow w}^* \preceq \Gamma \\
\iff & (P_{11} \mathbf{\Phi} + P_{12} \mathbf{X}) \underbrace{\mathbf{\Phi}^L \mathbf{\Phi}^{L*}}_{(\mathbf{\Phi}^* \mathbf{\Phi})^L} (P_{11} \mathbf{\Phi} + P_{12} \mathbf{X})^* + \underbrace{(P_{11} \mathbf{\Psi})(P_{11} \mathbf{\Psi})^*}_{\Lambda} \preceq \Gamma \\
\iff & \begin{bmatrix} \Gamma - \Lambda & (P_{11} \mathbf{\Phi} + P_{12} \mathbf{X}) \\ (P_{11} \mathbf{\Phi} + P_{12} \mathbf{X})^* & \mathbf{\Phi}^* \mathbf{\Phi} \end{bmatrix} \preceq \Gamma \\
\iff & \begin{bmatrix} \Gamma - \Lambda & (P_{11} \mathbf{\Phi} + P_{12} \mathbf{X}) \\ (P_{11} \mathbf{\Phi} + P_{12} \mathbf{X})^* & \mathbf{\Phi}^* \mathbf{\Phi}_c + \mathbf{\Phi}_c^* \mathbf{\Phi} - \mathbf{\Phi}_c^* \mathbf{\Phi}_c \end{bmatrix} \preceq \Gamma
\end{aligned} \tag{22}$$

Where $\mathbf{\Phi}_c$ is computed from an existing stabilizing controller.

4.4 Communication delays

Up to this point, the optimized controller does not incorporate considerations for communication delays. We will initially outline how to introduce these delays into the optimization problem. Subsequently, we will explore methods to address uncertainties associated with these delays.

4.4.1 Fixed terms in the controller

To accommodate potential communication delays between the controlled nodes, fixed terms associated with the variable $\mathbf{X}$ can be introduced into the convex optimization problem. Subsequently, the problem constraints can be reformulated as follows:

$$\begin{bmatrix} \Gamma - \Lambda & (P_{11} \mathbf{\Phi} + P_{12} [\mathbf{X} \circ F_x]) \\ (P_{11} \mathbf{\Phi} + P_{12} [\mathbf{X} \circ F_x])^* & \mathbf{\Phi}^* \mathbf{\Phi}_c + \mathbf{\Phi}_c^* \mathbf{\Phi} - \mathbf{\Phi}_c^* \mathbf{\Phi}_c \end{bmatrix} \preceq \Gamma, \quad \forall \omega \in \Omega \tag{23}$$

Where F_x is defined with 1 on the diagonal and with delays on communication links.

$$F_x = \begin{bmatrix} \mathbf{1} & \mathbf{z}^{-\delta_1} & \mathbf{0} \\ \mathbf{z}^{-\delta_1} & \mathbf{1} & \mathbf{z}^{-\delta_2} \\ \mathbf{0} & \mathbf{z}^{-\delta_2} & \mathbf{1} \end{bmatrix} \quad (24)$$

4.4.2 Delay uncertainty

Moreover, in cases where the delays are uncertain and lie within a known uncertainty set $(\delta_1, \delta_2) \in \Delta$, this uncertainty can be treated as multimodel uncertainty. One approach to address this is to incorporate constraints for each state of the multimodel. Consequently, the constraints of the optimization problem are modified to:

$$\begin{bmatrix} \Gamma - \Lambda & (P_{11}\Phi + P_{12}[X \circ F_x]) \\ (P_{11}\Phi + P_{12}[X \circ F_x])^* & \Phi^*\Phi_c + \Phi_c^*\Phi - \Phi_c^*\Phi_c \end{bmatrix} \preceq \Gamma, \quad \forall \omega \in \Omega, \forall (\delta_1, \delta_2) \in \Delta \quad (25)$$

4.5 Results

4.5.1 Distributed Data-Driven controller

In this section, we will analyze the performance of the controller. As discussed earlier, the primary goal of the controller is to maintain the grid frequency around 50 Hz. Therefore, our objective is to minimize the settling time and overshoot of the frequency following a perturbation. We will compare the performance of the optimized data-driven controller with that of the droop controller. It's important to note that, in this initial experiment, communication delays are neglected.

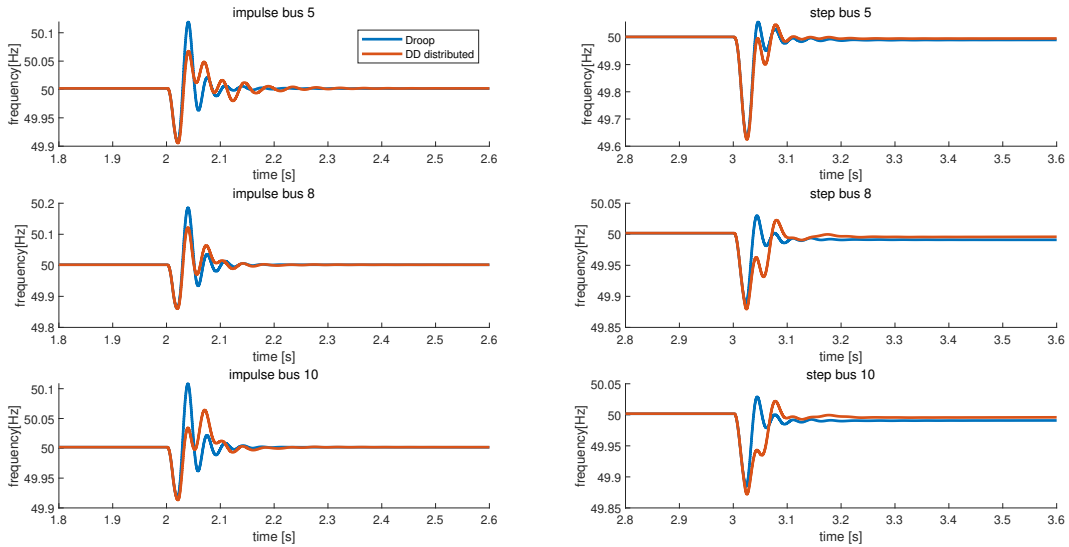


Figure 13: Comparison of frequency performances between the droop controller 'blue' and Data-Driven distributed controller 'red' after impulse 'left' and step perturbation 'right'

		Node 5		Node 8		Node 10	
		Droop	DD	Droop	DD	Droop	DD
impulse	settling time [s]	0.277	0.277	0.281	0.283	0.277	0.284
	overshoot [Hz]	0.119	0.094	0.185	0.139	0.108	0.086
step	settling time [s]	0.298	0.296	0.248	0.281	0.264	0.278
	overshoot [Hz]	0.368	0.371	0.112	0.116	0.115	0.122

Table 5: Distributed controller frequency performances

From the previous figure and the corresponding table, we observe similar performances in terms of frequency stability. The overshoot following an impulse perturbation on load 4 is noticeably reduced but remains quite similar. Now, let's turn our attention to the power-sharing performances. The results are presented in the figure below:

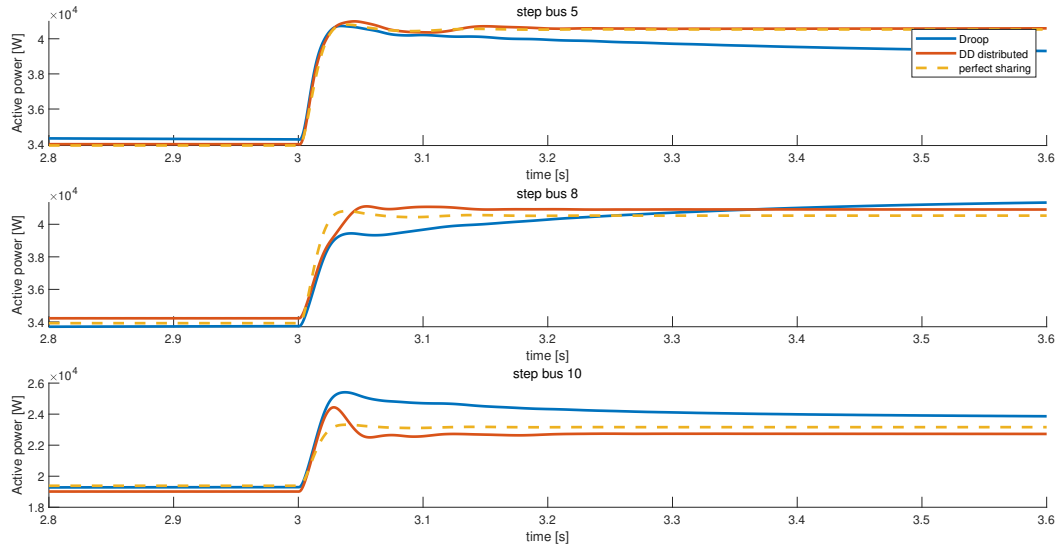


Figure 14: Comparison of power sharing performances between the droop controller 'blue' and Data-Driven distributed controller 'red'

		Node 5		Node 8		Node 10	
		Droop	DD	Droop	DD	Droop	DD
settling time [s]		2.088	0.269	2.218	0.231	1.298	0.278

Table 6: Distributed controller power sharing performances

The power-sharing performance is significantly improved by this controller, while the frequency stability remains within acceptable bounds.

4.5.2 Distributed Data-Driven controller with delay multimodel uncertainty

Now, our objective is to examine the differences in performance between a controller designed with assumed known delays ($\delta_1 = 1, \delta_2 = 2$) and a controller designed with multimodel uncertainty ($((\delta_1, \delta_2) \in$

Δ). We conduct a simulation test where $\delta_1 = 2$ and $\delta_2 = 1$. In this scenario, we assume that the first controller is incorrectly optimized.

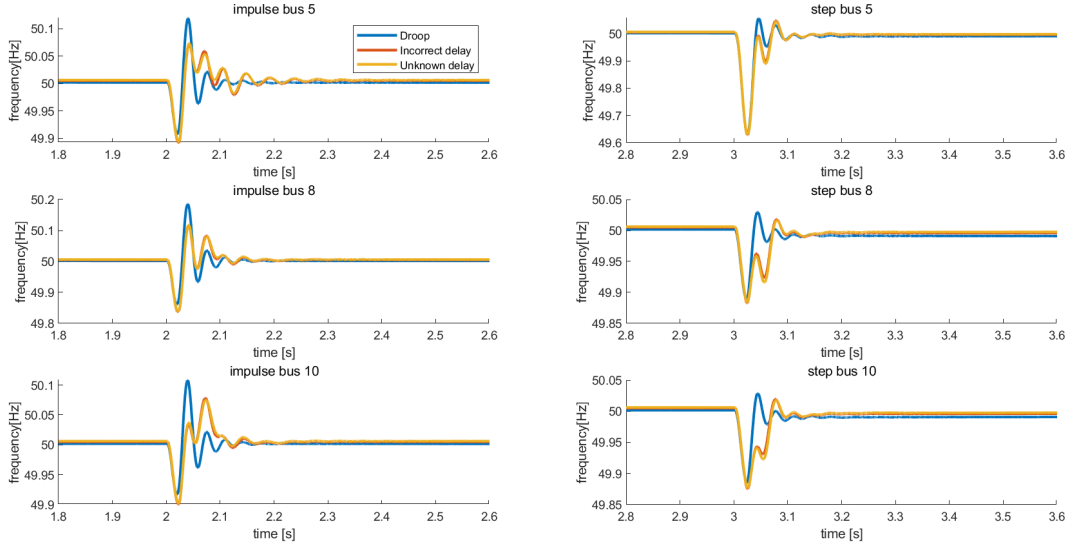


Figure 15: Comparison of frequency performances between the droop controller 'blue', incorrect delay Data-Driven controller 'red' and the multimodel delay data-driven controller

		Node 5		Node 8		Node 10	
		Incorrect	Unknown	Incorrect	Unknown	Incorrect	Unknown
impulse	settling time [s]	0.328	0.310	0.307	0.306	0.289	0.292
	overshoot [Hz]	0.108	0.107	0.163	0.161	0.100	0.098
step	settling time [s]	0.301	0.300	0.269	0.270	0.266	0.268
	overshoot [Hz]	0.372	0.371	0.117	0.116	0.125	0.124

Table 7: Multimodel controller frequency performances

As evident from the preceding figure and the corresponding table, the controller designed with uncertain delays exhibits superior performance in terms of grid frequency stability. The power-sharing performances are depicted in the following figure:

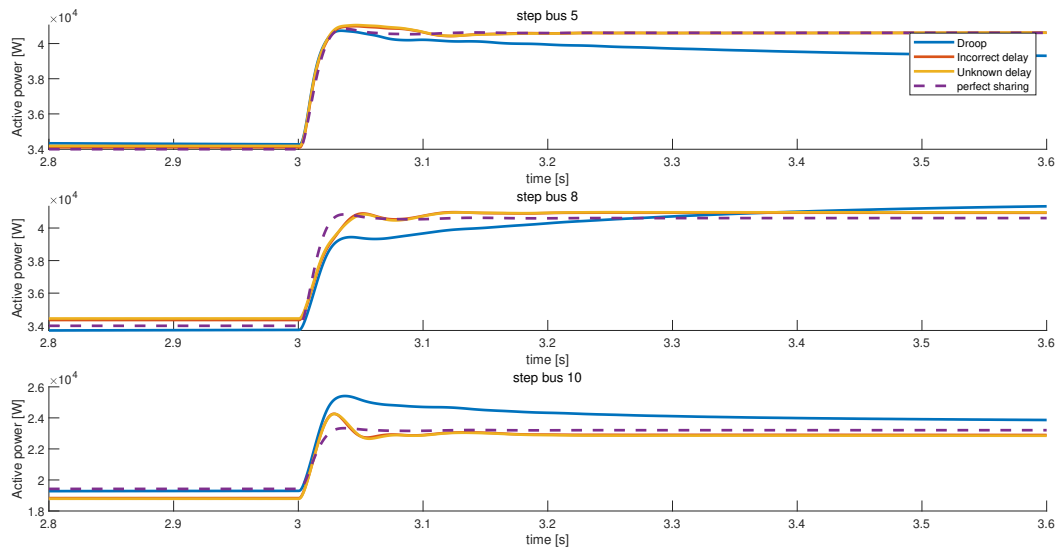


Figure 16: Comparison of power sharing performances between the droop controller 'blue', incorrect delay Data-Driven controller 'red' and the multimodel delay data-driven controller

	Node 5		Node 8		Node 10	
	Incorrect	Unknown	Incorrect	Unknown	Incorrect	Unknown
settling time [s]	0.215	0.216	2.238	0.239	0.255	0.262

Table 8: Multimodel controller power sharing performances

As indicated by the previous table, we notice slightly slower performance in power sharing for the controller designed with uncertain delays. However, it was previously discussed that the primary focus for this controller is to ensure frequency stability, which is indeed demonstrated by its superior performance in that regard.

5 Conclusion

In summary, this report aimed to introduce a methodology for designing a microgrid controller based on specific frequency and power-sharing specifications. While existing approaches employing a data-driven methodology have addressed this topic, none have adequately considered the uncertainty associated with delays between controlled nodes. To address this crucial aspect, we conducted a comprehensive simulation using Matlab and Simulink, re-implementing the entire experiment. Utilizing identification methods, we compared our simulation results with a linearized model, subsequently adopting the latter for the subsequent phases of the experiment.

The initial phase involved establishing a distributed controller without any delays. Subsequently, we proposed a novel approach employing multimodel uncertainty to account for the entire range of delay uncertainties. Constraints were strategically introduced into the optimization problem to ensure its resolution remained robust across all potential delays.

This multi-faceted approach not only enhances the robustness of the microgrid controller but also contributes to the broader understanding of control strategies in the context of uncertainty. As we advance towards more complex and dynamic energy systems, incorporating such methodologies becomes imperative for ensuring the resilience and adaptability of microgrid control solutions.

References

- [1] S. S. Madani, C. Kammer, and A. Karimi, “Data-driven distributed combined primary and secondary control in microgrids,” May 2021.
- [2] C. Kammer and A. Karimi, “Decentralized and distributed transient control for microgrids,” Jan. 2019.
- [3] P. Schuchert, V. Gupta, and A. Karimi, “Data-driven fixed-structure frequency-based h_2 and h_{∞} controller design,” Oct. 2023.